

Question Number	Answer	Additional guidance	Mark
4(a)	An answer that includes two of the following points: • minerals (1) • (suitable) pH (1) • (suitable) light (intensity) (1)	ACCEPT named mineral e.g. phosphate, nitrates ACCEPT oxygen, nitrogen, carbon dioxide, water ignore nutrients / moisture / predation / glucose / temperature ACCEPT suitable wavelength of light / sunlight	(2)

Question Number	Answer	Mark
4(b)(i)	The only correct answer is D William Withering <i>A is not correct because Withering tested his digitalis soup</i> <i>B is not correct because Withering tested his digitalis soup</i> <i>C is not correct because Withering tested his digitalis soup</i>	(1)

Question Number	Answer	Additional guidance	Mark
4(b)(ii)	An explanation that includes the following points: - patients given a range of {doses / concentrations} / doses increase in concentration (throughout the investigation) (1) (to allow selection of) lowest effective concentration / highest dose that can be tolerated without serious side effects (1)	ACCEPT (selection of) dose that is effective with minimal side effects	(2)

Question Number	Answer	Additional guidance	Mark
4(b)(iii)	A description that includes the following points: (selection of) patients with (colon / kidney) cancer (1) {half / one group} given {curacin A / (this) cancer drug} and {half / one group} given current (best) drug (1) - neither doctor nor patient knows which treatment has been given (1) - analysis of data to see if curacin A is more effective than the {current best drug / placebo} (1)	ignore ref to stage of trial ACCEPT placebo / sugar pill ACCEPT analysis to see if there is a {significant difference/improvement} between those who took the drug compared to the placebo	(4)